

**Syllabus
EGR 346- Machine Design (3 units)
California Baptist University
Gordon and Jill Bourns College of Engineering
Fall 2025**

Dr. Keith Hekman 951-343-4974 khekman@calbaptist.edu TEGR344	Tentative Office Hours: MWF: 9:00-10:30 AM R: 9:00-12:00 AM R: 1:30-3:00 PM Or by appointment
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Purpose of the course

This course is designed to develop your skills in the mechanics of materials to better design mechanical parts to avoid failure. Also, standards/guidelines for designing typical machine elements are covered.

Course Description (From CBU Catalog):

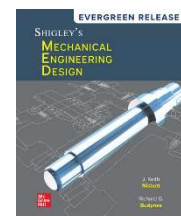
The fundamentals of machine elements in mechanical design. Includes the analysis of components under static and fatigue loadings, and the analysis, properties, and selection of machine elements such as shafts, gears, belts, chains, brakes, clutches, bearings, screw drives and fasteners.

Prerequisite:

EGR 242 Strengths of Materials.

Textbooks:

Shigley's Mechanical Engineering Design: 2024 Release by Budynas and Nisbett. The book is available on Blackboard via Immediate Access. Students can change their decision to opt in or out until September 16, 2025, 11:59 PM.





Course Description: EGR346 Machine Design The fundamentals of machine elements in mechanical design. Includes the analysis of components under static and fatigue loadings, and the analysis, properties, and selection of machine elements such as shafts, gears, belts, chains, brakes, clutches, bearings, screw drives and fasteners. Prerequisite: EGR242. (3 units; Fall)		Undergraduate Program Level Student Learning Outcomes	University Student Outcomes – Core Four*
Goal 1: Mechanics of Materials Knowledge			 
	CO1.1 Use the knowledge in Statics, Dynamics, and Strength of Materials for the design of machine elements. (CH 1-6 HW, test 1, final, Project 1)	#1, #2	
	CO 1.2: Use the concepts of failure theories and apply them in machine design. (CH 5-6, test 1, final, Project 1)	#1, #2	
Goal 2: Specific Machine Component Design			 
	CO 2.1: Design shafts for rotating machinery. (Chapter 7 HW, test 1)	#1, #2	
	CO 2.2: Select appropriate bearings, gears, belts, and screws for machine design. (Chapter 8,11,13,16 HW, test 2, final)	#1, #2	
Goal 3: Communication			
	CO 3.1: Communicate effectively through written skills. (Projects 1 and 2)	#3	

Undergraduate Program Outcomes related to Course:

- #1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
- #2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- #3. an ability to communicate effectively with a range of audiences



Class/Laboratory Schedule –

The class meets at the following times. Both sections are at about room capacity.

EGR 346A: MWF 10:45-11:45

EGR 346B: MWF 1:15-2:15

Time Expectations:

To be successful in this course (and in general in college) it requires spending 2 hours outside of class time for every hour in class.

Academic Integrity:

Discussion of assignments with other students is generally encouraged, unless specifically prohibited by the instructor. All work submitted for grading, however, must be your own or your groups, in the case of group work. Any evidence of plagiarism, academic dishonesty, or other violations of the CBU Honor Code may constitute grounds for a failing grade in the course. Use of homework solutions not provided by the instructor is prohibited. For further guidance please refer to the CBU Student Handbook for the CBU Honor Code and to the School of Engineering Policy on Academic Integrity.

Academic Environment:

Students are expected to encourage and maintain a positive learning environment that is reflective of a "Christian community of learners." Actions and behavior that distracts fellow students or is disruptive in the classroom will not be tolerated. To help maintain a positive learning environment, the following policies will be enforced:

- Cell phone use is prohibited in the classroom, unless specific permission is granted on a case-by-case basis. Cell phones can be used for audience polling. Cell phones should be turned to off or vibrate prior to class.
- Laptops should be closed, unless it is only used for class purposes (no email, TikTok, etc. during class)



- No food may be eaten in the classroom at any time, unless specific permission is granted before class on a case-by-case basis.
- Drinks (e.g., coffee, water, etc.) may be consumed during class as long as it is not disruptive of the learning environment. Drinks should not be consumed next to the laptops. Drinks should have lids.
- At the end of class, all students are expected to clean up after themselves. If moved, desks and classroom furniture are to be arranged as they were found at the beginning of class.

Clickers

TurningPoint (<https://participant.turningtechnologies.com/en/join>) will be used for audience participation and possibly for attendance. There is a link on Blackboard to link your account to Blackboard as well as a link for real-time polling.

Homework:

In order to facilitate learning, homework will be assigned weekly. Many of the homework problems that I have created are on Blackboard. These are homework problems similar to the textbook, but with different numbers. These will be done individually.

For the problems not on Blackboard, I will not assign HW groups, but you can work together in a group of up to 4 people if you want. You will turn in your homework on **Gradescope**. One person from the group will submit the assignment. With your submission, you need to indicate the other group members for them to receive credit.

If you start early enough, I recommend using the 10 minute rule for problems. If, after 10 minutes, you don't have any idea how to solve the problem, then ask me. There is a high probability if you don't know how to start a problem after 10 minutes, you will still not know after an hour.

Homework is due at the beginning of Wednesday classes and be based on the previous week's material.



The textbook's solution will be provided, since it is available for students who subscribe to Chegg or similar services. The textbook solutions are not always correct, and I am only providing them to make it equitable for all students. Because the solutions are provided, group homework must be completed in the format given in the homework guidelines. For some problems, I will ask for something different from what the book asks. If you blindly copy the textbook solution rather than answering the question I ask, you will get a maximum of half credit.

Grading Policy:

Attendance/Professionalism	5%
Assignments	10%
Projects	20%
Best Test	20%
Worst Test	15%
Final Exam	30%

Tests and Exams

- There will be two tests and a final exam.
- Make-ups for missed exams will be allowed in the event of an illness or emergency, but the instructor must be notified and approve it before the exam (in the case of an emergency, notify the instructor within 24 hours after the exam, and show an original note from a medical doctor, police, or other relevant agencies). There will be no make-ups for unexcused absences or absences notified after 24 hours of the exam time.

Generative AI Use

Generative AI use is prohibited on homework assignments. Generative AI use is encouraged on the project, but you must document how use used it. In other cases, the limitations of AI use will be specified.

Recording Class Sessions

I will record the class sections on Webex. Other recording of class sessions without the prior express written permission of the instructor is prohibited.



Any permission granted shall include the requirements that a recording may only be used for content study purposes only and sharing a recording with anyone outside of the course and/or posting on social media is strictly prohibited. This course policy is in alignment with the Student Handbook and the Standard of Student Conduct. Refer to Student Handbook policies 15.6, 15.7, and 15.8 for more information.

Grading Scale:

The following approximate grading scale will be used.

	93%-100% : A	90%-93% : A-
87%-90% : B+	83%-87% : B	80%-83% : B-
77%-80% : C+	73%-77% : C	70%-73% : C-
67%-70% : D+	63%-67% : D	60%-63% : D-
Below 60 : F		

FE Exam Content:

This course covers some of the basic information required for answering FE Exam questions for following sections (listed by discipline): Mechanical Engineering (FE Mechanical): Section 9, Mechanics of Materials Civil Engineering (FE Civil): Section 8, Mechanics of Materials Since the successful passing of the practice FE exam is required for the above disciplines, you are strongly encouraged to maintain materials (textbook, homework problems, lecture notes, etc.) for future use and place extra emphasis on studying practices that will help with retention of the presented information.

Contribution of Course to Meeting the Requirements of the ABET**Criterion 5:**

Estimated content: Engineering Science – 1 credits.
 Engineering Design – 2 credits.



Tentative Course Schedule

		Date	Chapter	Topic
1	W	9/3	1: Introduction to Mechanical Engineering Design	Introduction
2	F	9/5	1: Introduction to Mechanical Engineering Design	General Engineering Principles, Tolerances, and Design Factor
3	M	9/8	2: Materials	Material Strength and Stiffness, Hardness, Temperature Effects, Manufacturing, Steels, Heat Treatment, and Materials Selection
4	W	9/10	2: Materials	Material Strength and Stiffness, Hardness, Temperature Effects, Manufacturing, Steels, Heat Treatment, and Materials Selection
5	F	9/12	3: Load and Stress	Statics and Mechanics of Materials
6	M	9/15	3: Load and Stress	Statics and Mechanics of Materials
7	W	9/17	3: Load and Stress	Beams in Bending, Torsion, Stress Concentration, Pressure Vessels
8	F	9/19	3: Load and Stress	Beams in Bending, Torsion, Stress Concentration, Pressure Vessels
9	M	9/22	4: Deflection and Stiffness	Spring Rates, Beam Deflection, Superposition, Singularity Functions
10	W	9/24	4: Deflection and Stiffness	Spring Rates, Beam Deflection, Superposition, Singularity Functions
11	F	9/26	4: Deflection and Stiffness	Statically Indeterminate Problems, Compression Members, Long Columns, Columns with Eccentric Loading
12	M	9/29	4: Deflection and Stiffness	Statically Indeterminate Problems, Compression Members, Long Columns, Columns with Eccentric Loading
13	W	10/1	4: Deflection and Stiffness	Statically Indeterminate Problems, Compression Members, Long Columns, Columns with Eccentric Loading
14	F	10/3	5: Failures Resulting from Static Loading	Static Strength, Stress Concentration
15	M	10/6	5: Failures Resulting from Static Loading	Failure Theories, Ductile Materials
16	W	10/8	5: Failures Resulting from Static Loading	Brittle Materials



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17	F	10/10	5: Failures Resulting from Static Loading	Failure Criteria
18	M	10/13	Exam - Chapters 1-4	
19	W	10/15	5: Failures Resulting from Static Loading	Failure Criteria
20	F	10/17	6: Fatigue Failure Resulting from Variable Loading	Fatigue-Life, Stress-Life, Strain-Life, Linear- Elastic Fracture Mechanics Methods
21	M	10/20	6: Fatigue Failure Resulting from Variable Loading	Endurance Limit, Fatigue Strength, Stress Concentration
22	W	10/22	6: Fatigue Failure Resulting from Variable Loading	Torsional Fatigue, Combination of Loading Modes, Surface Fatigue Strength
23	F	10/24	13: Gears - General	Conjugate Action, Contact Ratio,
24	M	10/27	13: Gears - General	Conjugate Action, Contact Ratio, Interference
25	W	10/29	13: Gears - General	Bevel Gears, Helical Gears, Worm Gears, Gear Trains, Force Analysis
26	F	10/31	7: Shafts and Shaft Components	Shaft Materials, Layout and Design for Stress
27	M	11/3	7: Shafts and Shaft Components	Shaft Deflection and Critical Speeds
28	W	11/5	7: Shafts and Shaft Components	Keys, Fits, and Tolerances
29	F	11/7	11: Rolling-Contact Bearings	Bearing Load Life and Reliability, and Reliability vs Life
30	M	11/10	Exam - Chapters 5,6,13	
31	W	11/12	11: Rolling-Contact Bearings	Radial and Thrust Loading, Variable Loading, Ball and Cylindrical Roller
32	F	11/14	11: Rolling-Contact Bearings	Bearing Assessments, Lubrication, Mounting, and Enclosure
33	M	11/17	17: Flexible Mechanical Elements	Flat and Round Belts, V Belts
34	W	11/19	17: Flexible Mechanical Elements	Timing Belts, Roller Chain, Wire Rope, and Flexible Shafts
35	F	11/21	8: Screws, Fasteners, and the Design of Nonpermanent Joints	Power screws, Threaded Fasteners, and Joint Stiffness
		11/24	Thanksgiving Holiday	
		11/26	Thanksgiving Holiday	
		11/28	Thanksgiving Holiday	



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36	M	12/1	8: Screws, Fasteners, and the Design of Nonpermanent Joints	Bolt Strength, Tension Joints, Preload, Fatigue Loading
37	M	12/3	Review	

Final Exam

Section A: Wednesday, December 10, 9:30 - 11:30 am

Section B: Wednesday, December 10, 12:00 – 2:00 pm